Bearings

Year 4 Pt 2 · **Lesson Worksheet**

Worked Examples

Three-figure bearings: Write the bearing for a direction that is 35° measured clockwise from North, as a three-figure bearing.

Answer: A bearing is always written using three digits, so 35° is written as 035°.

Compass to bearing: What is the approximate three-figure bearing for South-East (SE)?

Answer: SE lies exactly halfway between South (180°) and East (090°), so its bearing is 135°.

Back bearings: A hiker walks from point A to point B on a bearing of 072°. Find the bearing of A from B (the back bearing).

Answer: Since 072° is less than 180°, add 180°: 072° + 180° = 252°. The back bearing of A from B is 252°.

Bearings Basics (15 questions)

1. What is the bearing of North? Give your answer as a three-figure bearing.
2. What is the bearing of East? Give your answer as a three-figure bearing.
3. Write 8° as a three-figure bearing.
4. Write 240° as a three-figure bearing. A) 24 B) 024 C) 240 D) 0240 E) 24.0
5. A ship sails on a bearing of 090°. In which compass direction is it travelling?
6. What is the bearing of South, as a three-figure bearing?
7. A plane flies on a bearing of 180°. In which compass direction is it travelling? A) North B) East C) South D) West E) North-East
8. List the 8 compass directions in clockwise order starting from North.
9. What is the approximate bearing of North-East (NE)?
10. A walker is told to head on a bearing of 315°. Which compass direction does this match? A) North-East B) South-East C) South-West D) North-West E) West
11. Select ALL that apply: which of the following are valid three-figure bearings? A) 045° B) 7° C) 199° D) 099° E) 450°
12. What is the bearing of West, as a three-figure bearing?
13. A hiker walks on a bearing of 090° and then turns to walk on a bearing of 270°. What is the relationship between these two directions?
14. Select ALL that apply: which of these bearings point in a generally southerly direction (between 090° and 270°)? A) 060° B) 150° C) 200° D) 300° E) 350°
15. A compass needle points on a bearing of 000°. Which compass direction is this? A) North B) East C) South D) West E) North-East

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Reading & Finding Bearings (15 questions)

1. A ship's radar shows a lighthouse at a bearing of 050° from the ship. Describe, using a compass direction, roughly where the lighthouse lies relative to the ship.

2. The bearing of town B from town A is 065°. What is the back bearing (the bearing of A from B)?

3. The bearing of a hill from a campsite is 142°. What is the back bearing of the campsite from the hill?

4. A bearing of 300° is given. What is the back bearing? A) 100° B) 110° C) 120° D) 130° E) 140°

5. A boat sails from harbour P to harbour Q on a bearing of 210°. What is the bearing from Q back to P?

6. Select ALL that apply: which of these pairs correctly show a bearing and its exact back bearing? A) 020° and 200° B) 095° and 185° C) 160° and 340° D) 250° and 070° E) 300° and 110°

7. A bearing of 075° is measured from North, turning clockwise. Through how many degrees would you turn, starting by facing North, to face along this bearing?

8. Two signposts show that Hilltown is on a bearing of 128° from Riverside. What direction, roughly (to the nearest compass point of N, NE, E, SE, S, SW, W, NW), is Hilltown from Riverside? A) East B) South-East C) South D) South-West E) West

9. A bearing of 190° is given. What is the back bearing?

10. A plane flies on a bearing of 048° from airport X to airport Y. Describe the back bearing in words (using a compass direction) and give its value as a three-figure bearing.

11. A bearing of 358° is given. What is the back bearing? A) 088° B) 178° C) 180° D) 182° E) 268°

12. Select ALL that apply: a ship travels from a buoy to a jetty on a bearing of 085°. Which statements about the return journey (jetty to buoy) are true? A) The back bearing is 265°. B) The back bearing is found by subtracting 180° from 085°. C) The back bearing is found by adding 180° to 085°. D) The return journey is on a bearing of less than 180°. E) 085° and 265° differ by exactly 180°.

13. A bearing from a ship to a buoy is measured as 212°. What is the bearing from the buoy back to the ship?

14. On a map, a church is due North-West of a school. Give the bearing of the church from the school.

15. A diagram shows two points, C and D. The bearing of D from C is 310°. State the bearing of C from D.

Bearings Challenge (15 questions)

1. A ship sails from port A on a bearing of 060° for 8 km to point B, then changes course to a bearing of 150° and sails a further 6 km to point C. If the ship then retraces its route back from C to B, what bearing does it sail on?

2. A walker sets off from base camp on a bearing of 072° and walks 5 km to a lake, then walks back to base camp. What bearing does the walker travel on for the return journey?

3. A plane flies from city P to city Q on a bearing of 128°, a distance of 240 km. If the plane then needs to fly from Q directly back to P, on what bearing should it fly?

4. Two ships leave the same port at the same time: Ship 1 on a bearing of 040°, Ship 2 on a bearing of 160°. What is the angle between their two courses? A) 100° B) 110° C) 120° D) 130° E) 140°

5. A hiker walks on a bearing of 095° for 4 km, then turns through 90° clockwise and walks a further 3 km. What bearing is the hiker now walking on?

6. Select ALL that apply: the bearing from a boat to a buoy is 162°. Which of these statements are true? A) The back bearing (buoy to boat) is 342°. B) The buoy lies roughly to the south of the boat. C) The back bearing is found by subtracting 180° from 162°. D) The boat lies roughly to the north of the buoy. E) The angle between the bearing and its back bearing is 180°.

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7. A surveyor measures the bearing from point X to point Y as 205°. Due to a measuring error, the true bearing is actually 15° more than this. What is the corrected bearing from X to Y, and what is its back bearing?

8. A ship sails 10 km on a bearing of 030° from point A to point B. A second ship sails from point A on a bearing of 300° for 7 km to point C. What is the angle ∠BAC (the angle at A between the two ships' courses)? A) 90° B) 100° C) 110° D) 120° E) 130°

9. A plane flies from airport A on a bearing of 085° to airport B, then from airport B on a bearing of 265° to airport C. What can you conclude about the positions of A and C?

10. A lighthouse keeper notes that a ship is on a bearing of 310° from the lighthouse. What bearing would the ship's captain record for the lighthouse, as seen from the ship?

11. Select ALL that apply: a plane's flight bearing from City M to City N is 078°. Which of the following are correct ways to describe the return bearing from N to M? A) 078° + 180° = 258° B) 078° − 180° = −102°, so add 360° to get 258° C) 258° D) 180° − 078° = 102° E) The return bearing and the outward bearing differ by 180°.

12. A ship sails away from a harbour on a bearing of 200°. It then turns around and heads back towards the harbour. A second ship leaves the harbour on the bearing that the first ship is now using to return. Explain why the second ship is NOT travelling in the same direction as the first ship's original outward journey.

13. A bearing of 142° is given. Find the back bearing, and state which of the 8 compass points the back bearing is closest to. A) North B) North-East C) South-West D) North-West E) South

14. A ship travels on a bearing of 085° for 12 km to point B, then on a bearing of 265° for 5 km to point C. Explain what this tells you about the position of C relative to the ship's original starting point, and find how much distance the ship could have saved by sailing directly from the start to C instead of travelling via B.

15. The bearing from a campsite to a waterfall is $2x°$, and the back bearing from the waterfall to the campsite is $(x + 200)°$. Form an equation using the fact that the back bearing is 180° more than the outward bearing, and solve for x.

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